

Name: K RAJ BHARATH K RAJ BHARATH | Reg. No: RA2511003011548

Unit – 5 : Session – 6 : SLO – 1 | Minimum Spanning Tree — Prim's Algorithm

■ I. Graph Reference

- Edges with weights: 4-3(5), 4-5(9), 4-1(1), 4-2(3), 5-1(4), 1-2(2), 3-2(3), 3-6(8), 2-6(7).

■ II. Fill in the Blanks

- The number of edges in a Minimum Spanning Tree for a graph with 6 vertices is 5.
- The first edge selected by Prim's algorithm starting from vertex 1 is 1-4 (weight 1).
- The edge with the next minimum weight connected to the MST is 1-2 (weight 2).
- The total weight of the MST obtained is 17.
- Prim's algorithm uses a VISITED SET (visited array / min-priority queue) to track the vertices already included in the MST.

■ III. True or False

- Prim's algorithm selects the highest weight edge available at each step — FALSE (minimum weight).
- The MST constructed for this graph will have 5 edges — TRUE ($n-1 = 5$).
- Prim's algorithm forms cycles while adding edges — FALSE (it only adds edges to unvisited vertices).
- The edge 3-6 (8) will be included in the MST if lower weight connections are available — FALSE
- (edge 2-6 with weight 7 is lower, so 3-6 is excluded).

■ IV. Short Answer Questions

- Prim's algorithm grows the MST from a starting vertex; at each step it picks the MINIMUM-WEIGHT edge that
- connects a vertex already in the MST to a vertex outside it.
- Prim's MST algorithm assumes an UNDIRECTED, connected, weighted graph; in directed graphs the notion of a
- spanning tree differs (arborescence), and negative weights break the greedy 'minimum edge' justification.
- A MIN-PRIORITY QUEUE (binary heap) makes Prim's algorithm more efficient than a simple array: $O(E \log V)$.

■ V. Apply Prim's Algorithm Starting from Vertex 1

- Step 1: MST = {1}. Min edge out: 1-4 (1). Select 1-4.
- Step 2: MST = {1,4}. Candidates: 4-2(3), 1-2(2), 4-3(5), 4-5(9), 5-1(4). Select 1-2 (2).
- Step 3: MST = {1,4,2}. Candidates: 2-3(3), 4-3(5), 2-6(7), 1-5(4). Select 2-3 (3).
- Step 4: MST = {1,4,2,3}. Candidates: 3-6(8), 2-6(7), 1-5(4). Select 1-5 (4).
- Step 5: MST = {1,4,2,3,5}. Candidates: 2-6(7), 3-6(8). Select 2-6 (7).
- Order of edges selected: 1-4, 1-2, 2-3, 1-5, 2-6.
- TOTAL WEIGHT OF MST = $1 + 2 + 3 + 4 + 7 = 17$.